EXPERIMENT NO. 04

AIM: Simulation of integrator circuit using Pspice.

PROBLEM STATEMENT: To design op - amp as an integrator and plot its input and output waveforms.

SOFTWARE REQUIRED: PSPICE software.

COMPONENT REQUIRED:

- uA741 operational amplifier
- Two resistors (R1 = 5 kΩ, R2= 10 kΩ, Rf = 100kΩ)
- Capacitor (Cf=0.125 uF)
- Input voltage source (Square wave(VPULSE))
- Two DC voltage sources (+15 V and –15 V)
- Ground terminals
- Voltage markers for input and output

THEORY:

A circuit in which the output voltage waveform is the integral of the input voltage waveform is the integrator. Such a circuit is obtained by using a basic inverting amplifier configuration if the feedback resistor R_f is replaced by a capacitor C_f. The expression for the output voltage is given as,

$$V_o = - (1/R_f C_1) \int V_i \, dt$$

Here the negative sign indicates that the output voltage is 180^0 out of phase with the input signal. Normally between f_a and f_b the circuit acts as an integrator. Generally, the value of $f_a < f_b$. The input signal will be integrated properly if the Time period T of the signal is larger than or equal to $R_f C_f$. That is, $T \geq R_f C_f$

The integrator is most commonly used in analog computers and ADC and signal-wave shaping circuits.

CIRCUIT DIAGRAM:

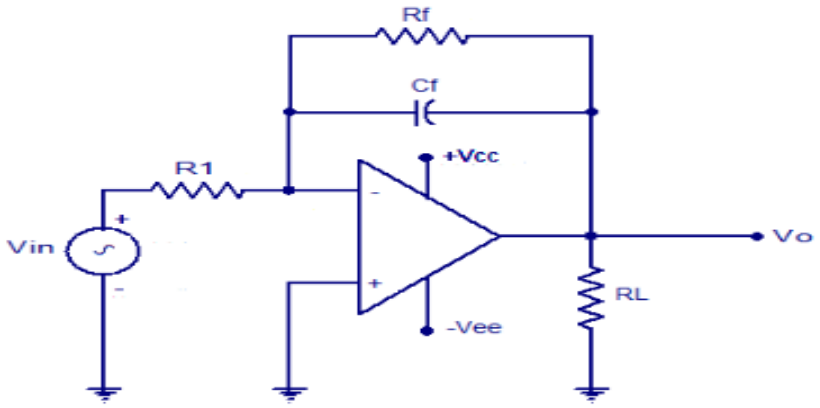


Fig 1: Integrator Circuit

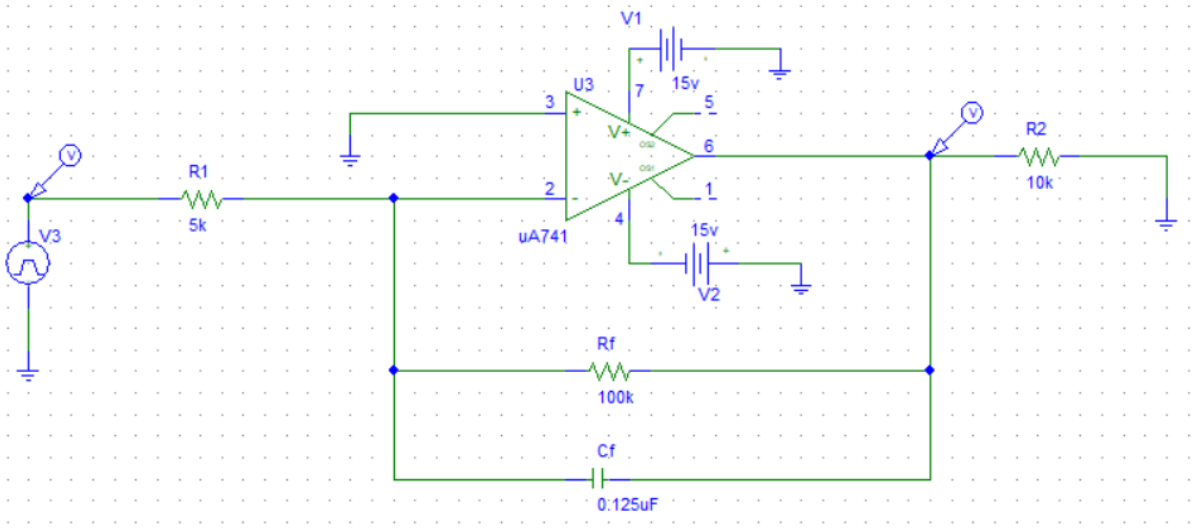


Fig 2: **:** Schematic of INTEGRATOR Circuit

PROCEDURE:

1. Design INTEGRATOR as per circuit diagram in Pspice Schematic.
2. Connect the non-inverting (+) terminal (Pin 3) of the uA741 to ground.
3. Connect the input signal through R1 to the inverting (–) terminal (Pin 2).
4. Connect Rf (100 kΩ) between the output (Pin 6) and the inverting input to provide feedback.
5. Apply +15 V to Pin 7 and –15 V to Pin 4 of the op-amp.
6. Set the component values:
    - R1 = 5 KΩ , R2=10 KΩ ,Rf=100 KΩ
    - Cf = 0.125 µF
7. Configure the VPULSE source:

    For a cleaner triangular output:
    - DC=0
    - AC=0
    - V1 = 5V
    - V2 = -5V
    - TD = 0 ns
    - TR = 1 ns
    - TF = 1ns
    - PW = 1 ms
    - PER = 2 ms
8. Create a **Transient Analysis** profile.
9. Set:
    - Print Setup = 10 us
    - Final Time = 70 ms

- No Print Delay= 50ms

10. Create **AC Sweep** profile.
    - AC Sweep Type = Linear
    - Total Point= 101
    - Start Frequency= 1 Hz
    - End Frequency = 1 KHz
11. Run the simulation.
12. Observe both input and output waveforms.

OBSERVATION:

S.No	Signal	Amplitude	Time Period
1.	Square wave input		
2.	Triangular wave output		

OUTPUT:

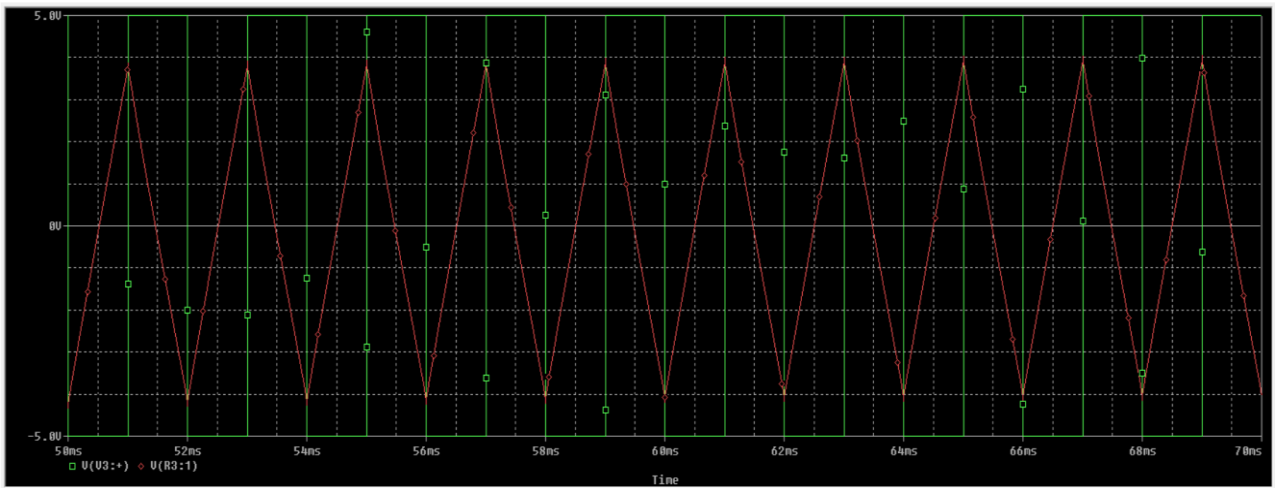


Fig : Integrator Output Waveform

RESULT:

The Op-Amp integrator circuit was successfully simulated in PSpice. A square-wave input produced a triangular-wave output, verifying the integrating action of the operational amplifier.

CONCLUSION:

VIVA VOCE:

1. What is an Op-Amp integrator?
2. Why is a capacitor used in the feedback path in integrator circuit?
3. Why is the output inverted in integrator circuit?
4. What happens if the capacitor is replaced with a resistor?
5. What are the applications of an integrator?